Section 11 Operator console

CONTENTS

1. INTRODUCTION

This section describes the backend items that are introduced with R7.1.

See also the WHAT's NEW paragraph below for more information.

Chapter 3 and up describe the R7.1-based hardware items

1.1. WHAT'S NEW

New features delivered with R7.1-1 with respect to R6.2:

Item	As delivered with R6.2	As delivered with R7.1-1
Reconstructor	SKY II	XP1000 6/500
Host CPU firmware version	V5.3	V5.4
Host Operating system	OpenVMS 6.2 -1H3	OpenVMS 7.1 - 1H2
Host SCSI controller IIM / ODR	KZPAA	KZPBA-CA
Print server	Emulex (optional)	HP Jetdirect 300X
Ethernet switch	Optional	5708TX
Console	Host expansion console	New furniture

Overview R7.1 console

This section gives an overview of the actions necessary to make the Operator Console operational.
The Operator Console consists of 3 units:

Operator Monitor Table:	Table for OCD monitor, Audio module, keyboard and mouse, printer and optional patient observation unit. The Service Storage Cabinet (SSC) is mounted beneath.
Operator Equipment Cabinet:	Contains the host computer and Backend Miscellaneous Box. This BE-box includes a DC Power supply, the IIM and the magneto-optical disk drive, the RS232 interface to the front-end, and the audio switch board.
Service Storage Cabinet:	Compartment for ethernet switch, print server, functional brain box and mains distribution unit (for operator equipment cabinet and miscellaneous equipment such as ethernet switch, print server, printer and optional modem).

The computer system is already installed by the MR factory in the Operator Equipment Cabinet.

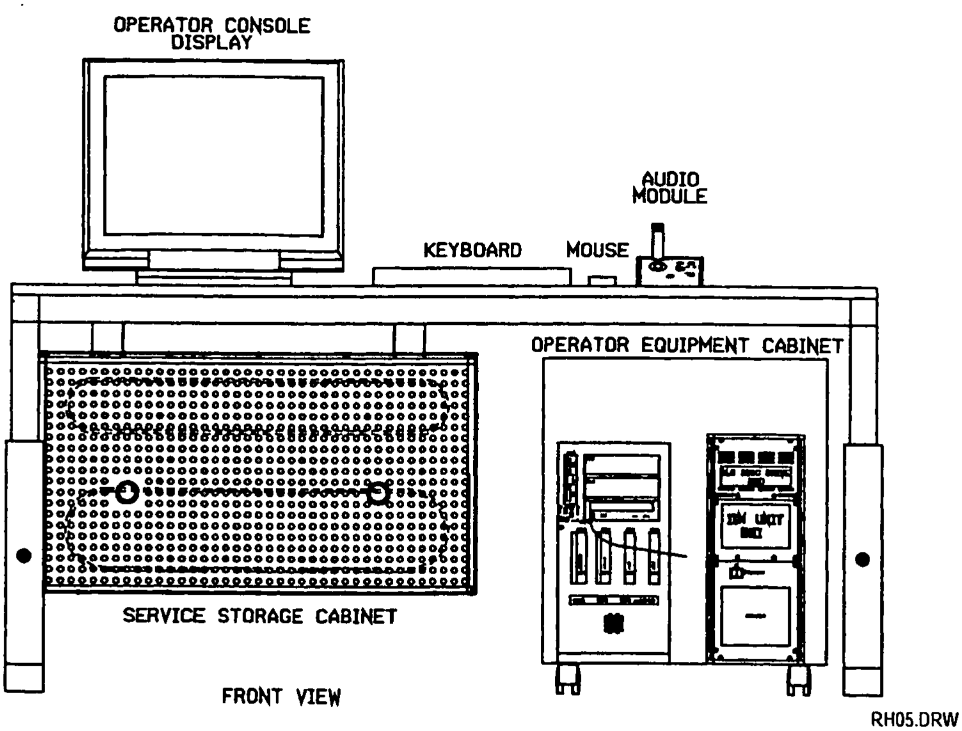


Figure 1: Operator console as delivered with R7.1

2. INSTALLATION

2.1. TOOLS AND TEST EQUIPMENT NEEDED

- Standard tool kit
- Multimeter

2.2. MAINS VOLTAGE SELECTIONS

The mains input voltage for the operator console is coming from the MDU in the equipment room.
For the Operator Console: always 230VAC. The voltage is connected to the mains distribution unit in the service storage cabinet.

NOTE

115 VAC outlets are not available in the Service Storage Cabinet.
A transformator 230VAC to 115VAC has to be purchased locally.

The following equipment is always connected to the 230 VAC output of the mains distribution rails in the service storage cabinet:

- LCD Operator Console Display
- Operator Equipment Console
- Ethernet switch
- Print server
- Printer
- Modem (option)
- Patient Observation Monitor (option)
- 230VAC to 115VAC transformator (to be purchased locally)

CAUTION

The 230 VAC outlets shall be used to connect to the above-mentioned equipment. This ensures that all equipment is connected to the same mains, thus preventing problems (such as IQ problems) caused by ground loops, spikes and dips on the mains voltage.

---

1. Connect the mains connectors to the local equipment.

2.3. INSTALLING OPTIONS

First install (if not already factory pre-installed) the video and/or the hardcopy options in the IIM in the console as described in Section 21.

2.4. INSTALLING THE NETWORK ITEMS

From R7.1 onwards, a network is part of the basic system.
The ethernet switch only supports UTP connections.
The ethernet switch and print server are placed in the service storage cabinet.
The ethernet switch can connect to 10Mb/s and to 100Mb/s links (autosense).

Because the ethernet ports are programmed, you must connect the units according Table 1. Refer to the drawings manual, section computer, for connector locations on the ethernet switch, drawing Z3-10.16.

Ethernet switch port number	Which unit
BNA-1X	Host computer
BNA-2X	Reconstructor (Located in NTDAC)
BNA-3X	Print server HP Jetdirect 300X
BNA-4X	Easy vision MR/CT workstation (optional)
BNA-5X	DICOM unit (optional)
BNA-6X	Spare
BNA-7X	Spare
BNA-8X straight (8MDI)	Hospital network straight (optional)

Table 1: Ethernet switch connections

1. Set the jumpers of the HP Jetdirect 300X print server according the drawings manual, section computer, drawing Z3-10.17.
2. Connect the mains connectors of the ethernet switch and print server to the mains distribution unit in the service storage cabinet.
3. Connect the RS232 cable (BHX16 – BNAX9) from the host computer to the ethernet switch.
4. Connect the following units at the ethernet switch according Table 1:
   - Host computer
   - Reconstructor
   - Print server

CAUTION

At this point, it is not allowed to connect the ethernet switch to the hospital network. Configuring and connecting the Gyroscan Intera to the hospital network is described in section 21.12 and can be started when the standalone Gyroscan Intera functions properly.

Testing of the network is described in section NTDAC and can only be executed after completion of the network configuration, which is described in this section and section NTDAC.

2.4.1. CONNECTING THE ETHERNET SWITCH TO THE HOSPITAL NETWORK (OPTIONAL)

As an option, it is possible to connect the ethernet switch to the hospital network.
Only one UTP outlet (straight) is supported.

CAUTION

IP network addresses from the host, reconstructor, switch, print server and other units connected to the ethernet switch must be changed when connecting the system to the hospital network!
It is not allowed to test the complete system with default network settings, connected to the hospital network.

Refer to section 21.12 for hospital network information and configuration.

2.5. CONNECTING KEYBOARD, MOUSE AND PRINTER

1. Connect the mouse and the keyboard to the extension cables for keyboard and mouse.
2. Plug the connectors at the rear of the AlphaServer 800 computer
3. Connect the printer cable from the printer to the print server parallel port.
4. Connect the printer DC-adapter to the mains distribution unit in the service storage cabinet.

2.6. CONNECTING THE OCD

1. Connect the mains cable to the mains distribution unit in the service storage cabinet.
2. Connect the video/sync cables coming from the video board inside the host computer to the OCD. Do NOT use the external 75 Ohm pieces on the Sync signals.
3. Connect the delivered RS232 cable coming from the serial input connector at the rear side of the OCD and connect it to BEBX5.
4. Fix the video cable and RS232 cable with cable-binders to the operator monitor table so that it does not have a force on the cables.
5. The drawings manual section Operator Console shows an overview of the rear side of the operator console display.

NOTE

A known problem with the OCD monitor is a black screen at switching the monitor ON. Solution is to switch OFF the OCD, remove all 5 sync/video cables from the OCD, switch ON the OCD, wait for the Stand-by text on the screen and now reconnect sync/video cables.

2.7. CONNECTING THE AUDIO MODULE

1. Connect the audio module to connector BEAX4 in the Operator Equipment Cabinet.

3. SETTING TO WORK OF THE HOST COMPUTER

3.1. ALPHASERVER 800 COMPUTER

NOTE

We assume that the mechanical installation of the Operator Console, described in previous paragraph, and the Hard Copy Unit (if available) have been done.

3.2. POWER-ON SELF TEST OF THE COMPUTER

1. At the MDU in the equipment room, switch on the mains supply to the operator console (circuit breaker F5).
2. Switch on the operator console display (mains switch at the rear of the monitor)
3. Switch on the host computer.
4. Set the mains switch of the Backend miscellaneous box in the Operator Equipment Cabinet in position "ON".

The AlphaServer 800 beeps once during a normal power-up.

The AlphaServer 800 performs a series of self-tests and startup routines. The firmware is initialised and the power-up selftest is performed. Results of the selftest are displayed on the monitor.

After successful completion of the self-tests the host computer will boot from the system disk.

3.3. ALPHASERVER 800 FIRMWARE UPDATE

In the field, the firmware levels of the flashROMs on the motherboard and on the KZPSA board can be updated. This should be done during initial software installation.
This is already done in the MR Factory with initial R7.1 deliveries.

The procedure to update the firmware is described in the Software Installation Manual (Formerly known as Release Bulletin)

3.4. PAPER PRINTER

The printer supplied with the MR system is used for printing the service files and for printing the spectro plots. From R7.1 onwards it is also possible to use the "paperprint" function. This creates a paper printout of clinical or phantom images on the printer.
The printer is connected to the print server. Configuration of the printer and print server is described in paragraph 4.3.

3.5. CD-ROM DRIVE

The CD-ROM drive in the AlphaServer 800 is default used as software load device. However it can also be configured as audio CD-player device. The audio CD-player function is optional in the Gyroscan.
This option can be ordered via the commercial catalogue. The cabling for the CD player is already prepared by the MR Factory. No hardware settings are needed. Both applications need their specific software driver.
Refer to the Software Installation Manual for the procedure.

4. SOFTWARE LOADING AND CONFIGURATION

From release 6.2-1 onwards the systems disk will contain the complete software already loaded.
Only site dependent configuration has to done:

- SW-key(s)
- Scanner hardware
- Network configuration
- Print server
- Options

Before you start configuring the software you will need the SW key(s) for this system. Check the system reference number and get the SW key(s) from the BBS.
The system will be configured as a Stand-alone Local Area Network without connection to the hospital network. You can test the complete system as a stand-alone system, and afterwards configure and physically connect it to the hospital network. This is described in section 21.12 of the SMI manual.

Refer to the following paragraphs in chapter 5 to complete the configuration.

The Software Installation Manual is used only in case of a complete initial software installation, or in case of a software upgrade.
We strongly recommend to study each new Software Installation Manual Besides the installation procedures for basic software and options, it informs you about

- New features/improvements/options
- Known problems and workarounds
- Using MRVMS
- Problem reporting
- Functions of user Gyrotool

Two floppy disks for the host are delivered with the Gyroscan system.
One floppy disk labelled “FSL/FW-Update V5.4 AS800” is meant as an ‘emergency rescue’ solution in case the firmware in the computer’s FlashROM is corrupt. Refer to the Decision Support System for the procedure

The floppy disk labelled “AS800 Syst Conf ECU” contains the drivers for the DIGI board. This floppy diskette is used to configure the DIGI board after a replacement. Refer to the Decision Support System for the replacement procedure.

4.1. CHECK FOR PRE-INSTALLED SOFTWARE

To check if the software has been correctly installed on the host computer, some checks have to be done.

1. Switch on the host computer and check if it will boot automatically
   If it does not boot, enter: >>>**sh auto_action** (parameter must indicate boot)
   To change this enter: >>>**set auto_action boot**
   Enter: .>>>**b**
2. Login under : **Gyrotest**
   Password: **gyrotest**
   `New password: <local choice>`
   `Verification: <local choice>`
3. Enter. **about** to check the SW release, version and level (CCS_release : ASWNT711)

NOTE

As the NTDAC is still switched off you will get a message: BSCHECKSW: Reconstructor is not reachable.

---

4. Check with the delivered software CD’s if this is the correct SW level installed.
5. Enter: **logout**

NOTE

If the host computer does not boot or does not have the correct SW level you will have to do an initial software installation.
Refer to the Software Installation Manual for the procedure.

4.2. CONFIGURE ETHERNET SWITCH

1. Login on the host computer with user **system**
2. Select: Configure Hardware Options
3. Select: Install Network switch
4. Select: Compaq Netelligent 5708 TX
5. Enter the MAC address of the switch:
   MAC-address (nn-nn-nn-nn-nn-nn/QUIT/?) : **nn-nn-nn-nn-nn-nn**

NOTE

The MAC address of the ethernet switch can been found on a label attached at the front, on the right lower corner of the switch.

---

6. Press **<Return>** after:
   Enter the host-name of the Switch [SWITCH]:
7. Press **<Return>** after:
   Enter internet address for SWITCH [192.192.192.212]:
8. The following switch will be installed:
   Switch type: Compaq Netelligent 5708 TX
   MAC-Address Compaq Netelligent 5708 TX: nn-nn-nn-nn-nn-nn
   Nodename Compaq Netelligent 5708 TX: SWITCH
   TCP/IP address Compaq Netelligent 5708TX: 192.192.192.212
9. Press **<Return>** to confirm the settings:
   Is this information correct? (YES/NO) [YES]:
   Application SW is stopped...
   Press Return for the main menu <**Return**>
10. Select: Exit to hardware options menu
11. Reboot the ethernet switch by switching the mains supply off and on again.
    Rebooting the ethernet switch takes 2 minutes.

4.3. CONFIGURE PRINT SERVER

1. Switch off the printer.
2. Login on the host computer with user **system** + <password>.
3. Select: Configure Hardware Options
4. Select: Install network printer (+ queue's)
   The system will display the '*configure printers*' menu.
5. Select: Supported HP deskjet printer
6. Select: **<paper size>** after:
   Enter paper size (A4/LETTER) [A4]
   (A4 is the European A4 format, LETTER is the US Quattro format).
7. Enter: <**MAC address**> (e.g. 00-60-B0-96-CC-87)

NOTE

The MAC address is written on a label, which is attached to the bottom of the printer server and is indicated with the text 'AD'. Refer to Figure 2.

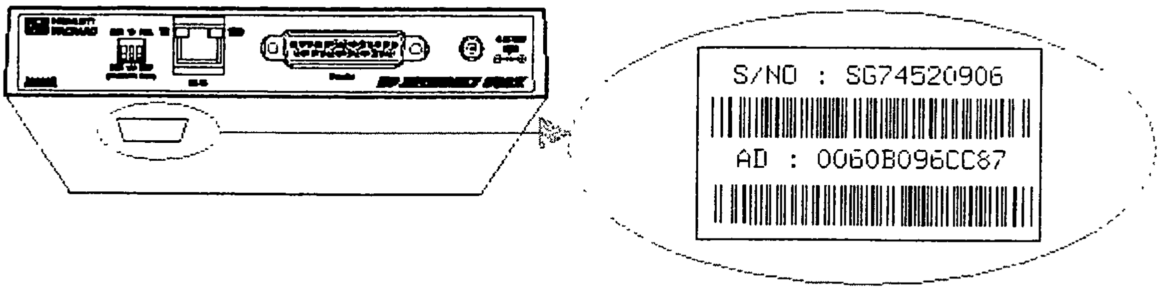


Figure 2: MAC address location print server

7. Press: <**Return**> after:
   `Enter the host-name of the JetDirect [JETDIRECT]:`
8. Enter the internet address for the *JetDirect*
   `Enter the TCP/IP address of the printer server:` **192.192.192.211**
9. Enter: **Yes** after:
   `Is this information correct? (YES/NO) [YES]`
10 The print server is now installed and the TCP/IP network is restarted.
11 Press <**Return**> after:
   `Press [RETURN] for main menu`

NOTE

Switch the power of the print server off and on.

12. **Wait 2 minutes for the print server to reboot.**
13. Switch on the power of the printer.
14. Printing a test-page can test the printer connection.
    If you want to test the printer, select option 3 Print test-plot. There are 2 test-prints available: a graphical test-plot (PCL) and an ASCII test plot.
    The ASCII test-plot suffices for the printer connection test
15. Select: `Exit to Hardware option menu`
16. Select: `Exit to Gyroscan manager Menu`
17. Select: `Logout` and confirm by clicking on **Yes.**

4.4. GYROTOOL CONFIGURATION

Refer to Appendix C: Scanner configuration, to get an overview of the possible configurations

1. Login on the host computer with user **Gyrotool** and press **<Return>**
2 Select: Modify system configuration
3 Select: System parameters.
4. If you want the change the language from English (default) to German:
   - Select: Language
   - Select: Deutsch
5. If you want to change the dataformat:

- Select: Date registration format.
- Enter the value.
- Select: **<Cancel>**

6. Select: Installation
7. Select: Hospital name
8. Enter the name (max. 30 characters)
9. Select: Service password
10. Enter a value (minimal 4 characters)
11. Select: System password
12. Enter a value (minimal 4 characters) and select **<Cancel>**
13. Select: Update locktype
14. Select: 99711
15. Select: Options (SW-keys).
16. Select: display/register permanent key. (Check here if the SRN is the correct one, see also the SRN number on the sticker, which must be placed on the MDU. If it is not the same, you have to load the complete software according to the initial procedure, refer to the Software Installation Manual.
17. Select: **<F14>** to erase the permanent key-field
18. Enter the permanent key in this field
19. Select: **<Proceed>** to validate the key
20. If you have temporary SW keys, they must be entered too:
    - Select: display/register temporary keys.
    - Select: **<F14>** to erase the temporary key-field
    - Enter the temporary key in this field
    - Select: **<Proceed>** to validate the key
    - Select: **<Cancel>** to leave Options menu
21. Select: Coil administration.
22. Enter the local coil configuration. For every coil present fill in the NT parameter. Only for Intera 0.5 T and for Intera 1.5 T Omni there are some exceptions, see Appendix C: Scanner configuration.
23. Select: **<Proceed>** to confirm the coil configuration.
24. Select: Console configuration.
25. Select: Graphics printer driver.
26. Enter the values and select: **<Proceed>**
27. Select: Copy facilities. (for more information on protocol selection press the "i" help button or refer to section 21.4 Hardcopy option)
28. Enter the values and select: **<Proceed>**
29. Select: Image interface module.
30. Enter the values and select: **<Proceed>** See section 21.11 for more details.
31. Select: **<Cancel>**
32. Select: Scanner Hardware Configuration.
    Adjust the appropriate entries (for more information, use "i" help button). Refer to Appendix C: Scanner configuration.
33. Select: **<Proceed>** when you are ready.
    In case of a message ("hardware configuration modified. Restart scanner by entering gyrorestart"), you must press **<Proceed>** again.
34. Select: Exit
35. Select: Exit from Gyrotool

4.5. TSW CONFIGURATION

1. Login on the host computer with user **TSW**
2. An automatic update of most items in the TSW Hardware configuration file is performed with the automatic update command at first login.
   It is still advised to inspect and if necessary manually edit the file.
3. Enter : **Conf**
4. Press : **<Return>** to Inspect
5. Check the file contents. Especially check the following line:

826 IIMOBOARD1 type =IIM_NONE **Check for correct video board if this option is present**

6. Press: **<Return>** to continue reading the file
7. Enter. **Edit**
8. Fill in your name and system id. and reason for editing

The following procedure can be used to manually edit the file
Change the lines which are not updated automatically.

1. Enter: **<Ctrl> Z**
2. Enter. **Exit**
3. Press: **<Return>** to quit

4.6. SAR LEVEL SETTING

Only for systems in Japan:

1. Login: **Gyrotest** (+ password)
2. Enter. **$startacq**
3. Select: Scan control
4. Select: Scan Utilities
5. Select: Service mode
6. Select: Control parameters
7. Select: General control parameters
8. Select: SAR level alert
9. Enter **0.4** instead of 0.0
10. Select: <**Store**> soft button (This will permanently store this parameter)
11. Enter: **yes** after: Do you want to modify the Current control parameter [No,Yes]?
12. Enter: **SAR level 0.4 W/kg** after: Reason?
13. Press: <**Return**> 2x
14. Select: **<Proceed>**
15. Select: **<Cancel>**
16. Select: Exit
17. Enter: **logout**

4.7. PROTOCOL FILTERING

NOTE

Do this only when all options (keys) and coils are set correctly under GYROTOOL.

---

1. Login under **gyrotest (+password)** on the host
2. Enter: **copy gyro$site_origin:ppdbpt#000.idx gyro$site:ppdb.idx**
   Where: # is 1 in case of a Intera Standard system
   # is 2 in case of a Intera Omni system
   # is 3 in case of a Intera Power system
   # is 6 in case of a Intera Master system
3. Enter: **@gyro$com:bsppinstall**
4. Enter: **Yes** after:
   `BSPPINSTALL: Do you want to continue? (Y/N) [N]:`

The protocols will be filtered now, this will take at least 10 minutes.
The process is executed in background.

Check on a regular base with **show entry** until the message appears: no such entry or wait for the message : "Job PPINSTALL (queue SYS$BATCH, entry <nr>) completed"
The filtering is finished now.

5. After the batch-job has finished, check for correctness of the filtering:
   Enter: **type/page ppinstall.log**
   No error may be present.
6. When ready, enter: **logout**, double-click on icon MANAGERNTSCAN, click on **SESSION-END**, click on **End session** and confirm by clicking on **Yes**.

5. SETTING TO WORK OF THE AUDIO MODULE

The "Patcom" process on the host computer controls the functions of the audio module. The functions related to the audio module and audio can be tested with the TSW.

5.1. TESTING THE AUDIO CHAIN

1. On the host computer, login with user **TSW**.
2. Enter: **TPATCOM**
3. Enter: **RUN FUNCTIONAL INTERACTIVE**
   This function tests the complete intercom circuit, including the headset (if available).
   No errors are allowed.
4. Enter: **QUIT**

6. SETTING TO WORK OF THE OC DISPLAY

6.1. INTRODUCTION

After the complete software has been configured, the Operator Console Display (OCD) must be adjusted or checked. This adjustment should be done when the OCD has been placed on its final position. The OCD for the operator console is a high resolution (SXGA), multisync, autovoltage LCD display.

The OCD is connected to the host computer via RS232. The RS232 port on the display is used for:

- Blacklevel adjustment and default calibration settings.
- Switching to sleepmode after 20 minutes of no activity on mouse nor keyboard.
- (Un)-Lock the display OSD buttons.
- Retracting information from the OCD like operating hours, settings, H and V frequencies, firmware levels, etc.

The following OCD adjustments must be performed during installation:

- Blacklevel adjustment.
- Horizontal size and position.
- Vertical size and position.
- Phase tuning for a stable display.

6.2. SAFETY INSTRUCTIONS

CAUTION

Place the display on a flat and stable surface that can bear the weight of at least 3 displays. If you use an unstable cart or stand, the set may fall, causing serious injury to a person, and serious damage to the equipment.

CAUTION

Avoid reflections in the display screen to reduce eyestrain.
Avoid touching the display screen with fingers, as it is not hard glass surface.
The pressure can damage the crystals.
Do not cover or block the ventilation openings in the cover of the display. When installing the apparatus in a cupboard or another closed location, take care of the necessary space between the display and the sides of the cupboard. Keep the display away from heat sources.

6.3. POWER MANAGEMENT

The OCD makes use of a sleep mode, which is initiated by the host computer. When there are no activities on the host keyboard or mouse for 20 minutes, the display will be switched to the sleep mode. The backlights will be dimmed, but are not switched off completely. After another 40 minutes of inactivity the LCD will switch itself OFF.

NOTE

Do NOT switch off the OCD with the power switch.
Switching On/Off frequently will decrease the life-time of the backlights of the LCD

6.4. ADJUSTING THE OC DISPLAY

6.4.1. BLACKLEVEL ADJUSTMENT

The blacklevel setting is controlled via the host computer. This is necessary because the picture on the display and the image from the Hard Copy Unit will be calibrated to a certain similarity. This will never be 100% the same but it has to be adjusted to a customer satisfactory result.
Before starting the adjustments make sure that the ambient light is the same as when the operators are working with the system. The display must be switched on for at least 30 minutes before adjusting it.

1. Login with user **TSW** on the host.
2. Enter: **TMONITOR**

A pop-up window appears to control the DISPLAY adjustments, refer to Figure 3.

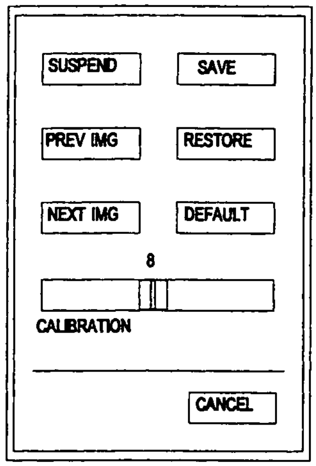


Figure 3: TSW TMONITOR screen

SUSPEND	soft button: enables/disables the OSD buttons at the rear side
DEFAULT	soft button: restores to factory settings
SAVE	soft button: saves current calibration setting
RESTORE	soft button: restores the previously saved settings
NEXT IMG	soft button: used to cycle forwards through test images
PREV IMG	soft button: used to cycle backwards through test images
CANCEL	soft button: cancels the operation
CALIBRATION	slider to adjust Blacklevel

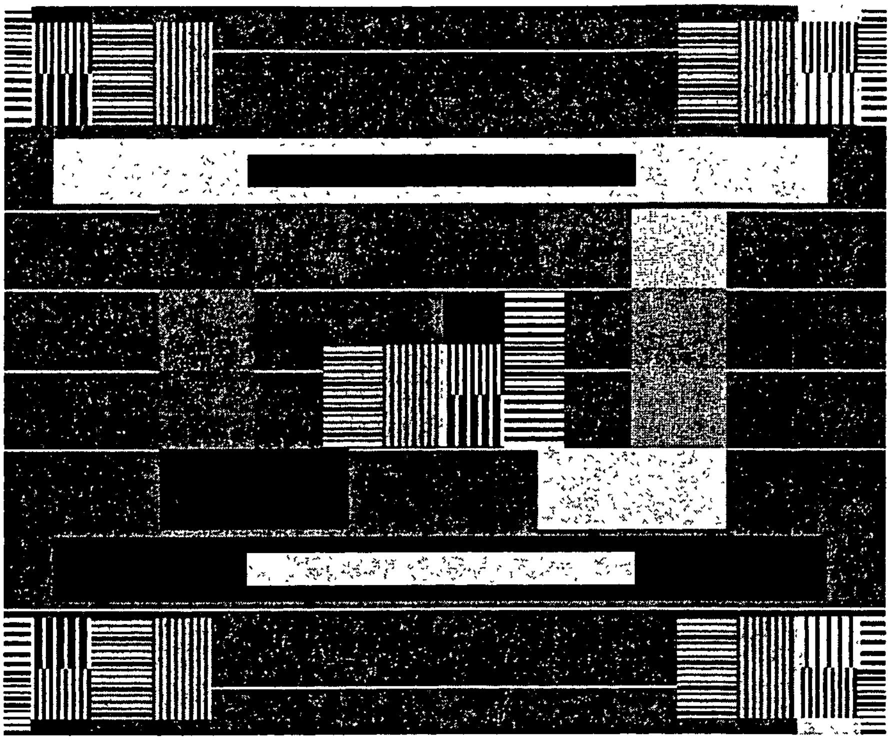


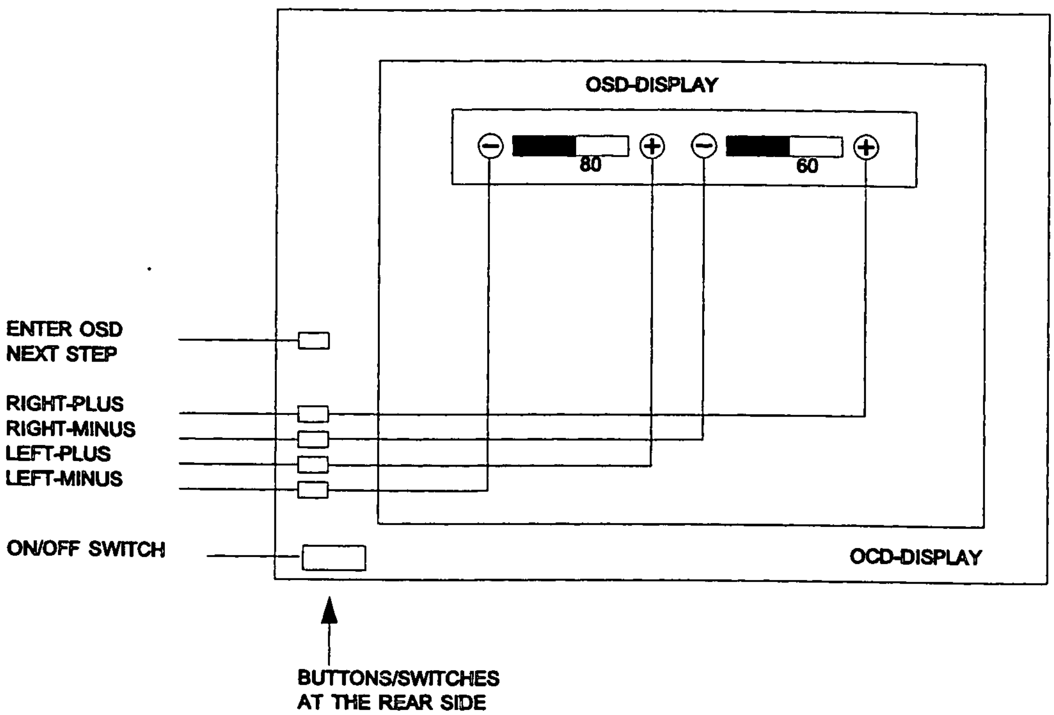
Figure 4: SMPTE pattern

3. The 5% and 95% video patches should be visible. The display is adjusted in the factory for 120 Cd/m^2, this is the default setting (value 5 on the calibration slider). If required, the blacklevel of the display can be adjusted by sliding (with the mouse) the calibration slider in the pop-up menu. However it is desired to keep this value as low as possible for a longer lifetime of the LCD backlights.
4. Select: <**SAVE**> on the pop-up menu.
   The present setting of the display is saved into a file. Application software uses this file to do the correct setting of the display when Gyroscan is starting.
5. Select: <**NEXT IMAG**> on the pop-up menu until you see the 0/5 % and the 95/100 % picture. Verify the visibility of the 5% and 95% section on this image.
6. Continue with the Phase adjustment

6.4.2. Phase adjustment

1. Select: <**NEXT IMAG**> on the pop-up menu until you see the POPO image.
2. Select: <**SUSPEND**>
3. Press on the display rear panel the "**Enter OSD**" button to get the On Screen Display menu. A contrast and brightness slider will be shown. Do not adjust these sliders.
4. Press on the display rear panel the "**Next step**" button to go to the next adjustment. An autoadjust selection will be displayed.
5. Press the "**Left-minus**" button and hold it until a message "Autoadjusting image. . . " is displayed to start the auto-adjustment. This selection will auto adjust the hor. and vert. positions and sizes. After the adjustment the horizontal size and position sliders will be shown. Do a manual fine-tuning on the Hor position with the "**Right-plus**" and "**Right-minus**" buttons. A uniform image without vertical bands should be the result.
6. Press on the display rear panel the upper "**Next step**" button. A Tune and Black slider will be shown

7. Optimize the POPO image until it is sharp and stable with the Tune scroll bar.
   Start at a setting of 25 and increase the tune value step-by-step with the "**Left-plus**" button until a stable and darker image is visible. Then decrease the setting step-by-step with the "**Left-minus**" button until again a stable and darker image is shown. Add the two values and devide by two to find the optimum setting (e.g. (18+30)/2 = 24). Place the tune slider on this optimum setting.
8. Press on the display rear panel the "**Next step**" button to go to the next menu entry. A vertical size and position slider will be displayed. No adjustments will be necessary.
9. Press on the display rear panel the "**Next step**" button to go to the next menu entry. A store and Advanced selection will be displayed. It is not necessary to store the values as the display saves the last adjusted settings
10. Press on the display rear panel the "**Next step**" button to go to the next menu entry. A timing view will be displayed. Check here if the frame buffer board inside the computer has been set on the correct vertical frequency.
11. Press on the display rear panel the "**Next step**" button to go to the next menu entry. A part info view will be displayed.
12. Press on the display rear panel the "**Next-step**" button to leave the OSD.
13. Select: **<SUSPEND>** on the pop-up menu, to lock the rear buttons.
14. Select: **<CANCEL>** on the pop-up menu, to exit the test.
15. Enter: **LOGO**
16. Select: **<YES>** to confirm the logout



FK68.WPG

Figure 5 : OCD button functions

7. APPENDIX A: VIDEO TIMING SPECIFICATION FOR THE OCD DISPLAY

Horizontal frequency · 63.3 kHz Range: 29 . 82 kHz
Vertical frequency: 60 Hz Range: 50 .. 76 Hz

Scan parameters	Pixels / Lines	Interval
Horizontal scan line	1708 pixels	15.79 microsec.
Displayed pixels/line	1280	11.83
Blanked pixels/line	428	3.96
Horizontal front porch	49	0.41
Horizontal sync pulse	204	1 70
Horizontal back porch	222	1.85
Vertical lines/frame	1056 lines	16 76 milisec
Displayed lines/frame	1024	16.17
Blanked lines/frame	32	0.51
Vertical front porch	3	0.047
Vertical sync pulse	3	0.047
Vertical back porch	26	0 41
Clock frequency		108.21 MHz

Picture dimensions (mm) : (399+/- 2) x (319+/-2)
Aspect ratio. 5 : 4
Display resolution: 1280 x 1024
Dimensions monitor (mm): height: 473, width: 507, depth: 225.
Power supply: 100-120/220-240 Veff (AC) - 50/60Hz automatic selection
Power consumption: 70 W
Power consumption stand-by: 5 W

OSD settings:
Via the OSD menu the settings of the monitor can be read.

Timing:
V: 60 Hz H: 63.4 kHz
SXGA H/V sync
V sync: NEG H sync: NEG

Next menu:
Part: 4522 131 59341
Serial: xxxxx
Software: P270P11B
Temp: Max xxx Actual xxx
Backlight: BLS xxx BLL xxx

Where:
Max. is the maximum measured internal temperature
Act. is the actual internal monitor temperature
BLS is the Backlight lamp driver output setting
BLL is the Backlight lamp luminance

During the lifetime of the backlights inside the display the BLS setting will slowly increase automatically, until a maximum setting of 255 will be reached. At this point the BLL setting which setpoint normally is 100 %, will decrease slowly. The luminance of the display will decrease slowly also. These settings can be read remotely via TSW, TMONINFO, run tools information console.

8. APPENDIX B: REPLACEMENT PROCEDURES HOST COMPUTER

8.1. ACCESSING THE FRU'S

WARNING

Before removing the top cover and side panels:

- *Perform an orderly shutdown of the entire system.*
- *Switch off the entire system.*
- *Disconnect all cables from the host computer.*
- *Unplug the AC power cord as last cable.*

CAUTION

Static electricity can damage integrated circuits. Always use the static protective service kit when working with internal parts of a computer system.

The computer's power supply does not automatically sense the input voltage. If you (plan to) test the computer stand alone (outside the operator console), ensure that the voltage switch is set correctly to match the mains voltage 115 VAC. or 230 VAC

After opening the computer cabinet and before closing it, check that the sticky fingers (metal strips to shield RF radiation) arė still in the side panels and that they did not come loose. Risk of short-circuit if they have come loose.

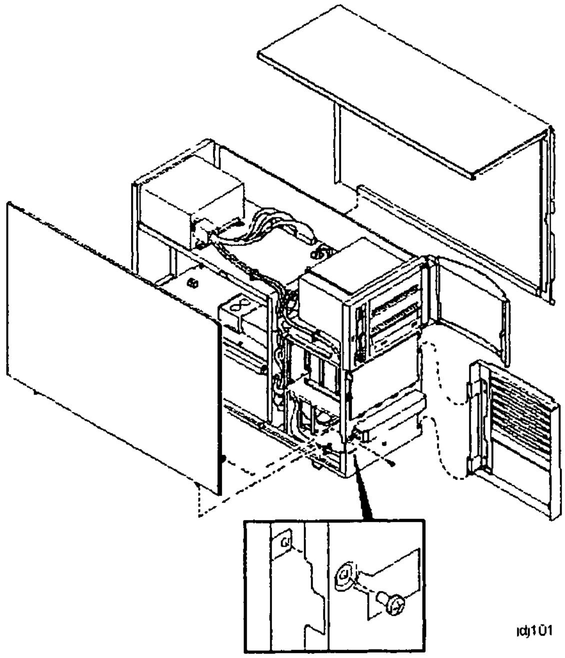


Figure 6: Removing the covers from the computer

1. Unlock the retaining screws from the brackets, which are used to mount the computer into the operator equipment cabinet. (2 screws at the front and rear)
2. Pull the computer towards you and remove the computer from the operator equipment cabinet.
3. Unlock and open the front door.
4. Remove the retaining screw indicated by the yellow label on the lower left side of the front of the system. See Figure 6.
5. Slide back and remove the left side panel.
6. Slide back and remove the top and right side panel.

Unless otherwise specified, you can install a field-replaceable unit (FRU) by reversing the steps shown in the removal procedure.

NOTE

The drawings manual, section computer contains the jumper settings of the boards and units.

---

1. Check the jumper settings of each new board. The factory settings are described in the drawing manual under section COMPUTER.
2. Whenever you add, remove or install an ISA or EISA board board which is connected to the EISA bus in the AlphaServer 800 computer, you need to run a utility called EISA Configuration Utility (ECU).
   Each EISA or ISA board has a corresponding configuration file, which describes the characteristics and required resources for that board. You need to run ECU to prevent conflicts on the EISA bus. If you have a bus conflict in the AlphaServer 800 computer then the power-up selftest does not pass, and the monitor indicates **ec** .The configuration files are stored in NVRAM (on the motherboard).

3. Run ECU if you swap or install an EISA board (i.e. the DIGI board in the AlphaServer 800) or NVRAM. ECU is described within their replacement procedures.
4. In addition, some boards contain an EPROM with firmware. The firmware level may also be updated when replacing the board. The board's firmware level is MR software release dependent.
5. If you replace one of the PC boards mentioned in table Table 2, then you have to update the firmware on these boards and/or the configuration file of the computer, respectively.

PC board to be replaced	Check/update firmware on board Y/N	Run ECU utility Y/N	Other remarks
CPU board	Y	N	
Frame buffer board	N	N	
KZPBA-CA board	N	N	Configure according to Software Installation Manual
KZPSA board	Y	N	Configure according to Software Installation Manual
DE500 board	N	N	
DIGI board	N	Y	Run ECU if TWRAP fails
NVRAM	N	Y	
Motherboard (backplane).	N	N	Swap old NVRAM and NVRAM TOY

Table 2 : Overview boards AlphaServer 800

For this purpose you will find some floppy disks and CD-ROMs from Compaq, which have been shipped with the MR system or the PC boards. These media contain the software for updating the firmware. Use the floppy disk drive and the CD-ROM from the AlphaServer 800 computer.

9. APPENDIX C: SCANNER CONFIGURATION

Configuration for initial systems

HW configuration	Intera 0 5 T	Intera 1 0 T		Intera 1 5 T			
Parameter	Standard	Omni	Power	Omni	Power	Master	Remarks
RF amplifier type	MR5002	Ehrhorn_S2x_42		Ehrhorn_S2x_64			
MN RF amplifier type	Not applicable	Not applicable		xxxx			If phosphor option present enter MR5003, if not enter none
B1 field	Not applicable	Not applicable		low	high	high	
MN transceiver board	Not applicable	Not applicable		xxxx			If phosphor option present enter available, if not enter not available If spectro option present enter 'available', if not enter 'not available'
Auxillary connector	Not applicable	Not applicable		xxxx			If spectro option present enter 'available', if not enter 'not available'
Synergy boards	xxxx	xxxx		Available			If synergy coil option present enter 'available', if not enter 'not available'
Grad Coil Type	aircooled	aircooled	watercooled	aircooled	watercooled		
Grad Amplifier type	Copley_234_02	Copley_234P4	Copley_274S	Copley_234P4	Copley_274S	Copley_274D	Copley_234_02 for grad amplifier with 3 power modules per axis Copley_234P4 for grad amplifier with 4 power modules per axis
Magnet type	NT_RMMU	NT_RMMU		NT_RMMU			NT for NT magnets with 36 shim rails NT_RMMU for magnets with fixed horizontal cold head
Acquisition system	Plus	Plus		Plus			Plus for BDAS system with "BCP plus" board (FCBX53 is a 10-pins JTAG connector)
BDAS backpanel 32 bit	Available	Available		Available			Not available for BDAS rack 4522 130 89002 or lower Available for BDAS rack 4522 130 89003 or higher
Patient support	MT	MT		MT			MT_Patsup for patient support with two table-top releases buttons
ECG option	Available	Available		Available			Always to be set to available
PPU option	Available	Available		Available			Always to be set to available
Respiratory option	Available	Available		Available			Always to be set to available
Headset	xxxx	xxxx		xxxx			Not available for system without the headset option Available for system with the headset option
Audio Switch Version	Version2	Version2		Version2			Version1 see info text in Gyrotool Version2 see info text in Gyrotool
Fepsu Type	In_SFB	In_SFB		In_SFB			In_SFB From R6 2 onwards the frontend power supply unit is built in the System Filter Box

Coil configuration:	Intera 0 5 T	Intera 1 0 T		Intera 1 5 T			
Parameter	Standard	Omni	Power	Omni	Power	Master	Remarks
QBC	Standard	NT	NT	Standard	NT	NT	
QHC	Standard	NT	NT	NT	NT	NT	
Quad knee	Standard	NT	NT	NT	NT	NT	
Neck quad/lineair	Standard	NT	NT	NT	NT	NT	

Note: All other coils are NT coils.